

BLUESTOCK FINTECH

Project

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Executive Summary

This report documents the end-to-end design and development of the Bluestock Fintech Mutual Fund Analytics Platform, a full-stack data engineering project built using Python, SQL, and Power BI. The platform ingests, cleans, stores, and analyses publicly available mutual fund data from AMFI India, delivering actionable insights through an interactive four-page dashboard.

The Indian mutual fund industry has witnessed explosive growth, crossing Rs. 81 lakh crore in Assets Under Management (AUM) across 1,908 schemes and 26.12 crore investor folios as of December 2025. Monthly Systematic Investment Plan (SIP) inflows reached an all-time high of Rs. 31,002 crore in December 2025, reflecting India's deepening equity culture and increasing retail investor participation.

This platform addresses five real business problems — data being scattered across multiple sources, no easy way to compare funds on risk-adjusted basis, no benchmark tracking, limited visibility into investor demographics, and slow static PDF reports. The solution brings everything into one place using an automated ETL pipeline, a normalised SQL database, and an interactive dashboard.

What this project delivers:

- Automated Python ETL pipeline processing 10 datasets and 87,000+ rows
- Cleaned and validated data stored in a SQLite database with a star schema design
- 15+ publication-quality charts from Exploratory Data Analysis
- Industry-standard risk and performance metrics — Sharpe Ratio, Sortino Ratio, Alpha, Beta, Maximum Drawdown, and Value at Risk
- A 4-page interactive Power BI dashboard with slicers, KPI cards, and drill-through

BUSINESS CONTEXT

Bluestock Fintech is a financial technology company that wants to make investment analytics easier for everyday investors in India. Most people who invest in mutual funds don't have access to proper tools that can help them compare funds, understand risks, or track their portfolio against a benchmark. They either rely on basic apps that show only raw returns, or they read static PDF reports that are weeks old by the time they see them.

This project builds the kind of data platform that solves exactly these problems — clean data, meaningful analysis, and a visual dashboard that anyone can use.

PROBLEM STATEMENT

Problem 1 — Data is Scattered Everywhere

NAV data, AUM data, SIP inflow data, and portfolio holdings are published on different pages of the AMFI website in different formats — some are TXT files, some are PDFs, some are HTML tables. There is no single unified database. This project solves it by building an ETL pipeline that collects everything into one SQLite database.

Problem 2 — No Easy Way to Compare Funds

If you want to compare SBI Bluechip vs HDFC Top 100 vs Axis Bluechip, you cannot just look at raw returns. A fund that returned 30% took on how much risk to do that? Without Sharpe Ratio, Alpha, and Beta, the comparison is meaningless. This project computes all these metrics for all 40 funds and presents them in a sortable scorecard.

Problem 3 — No Benchmark Tracking

Most retail investors have no idea whether their fund is actually beating the market or just riding the wave. Tracking error, Alpha, and Information Ratio all require the fund's NAV data to be compared against a benchmark index like Nifty 100. This project does exactly that using OLS regression.

Problem 4 — Limited Investor Behaviour Insights

AMCs and distributors don't have clear visibility into which age groups invest the most, which states contribute the highest SIP volumes, or how T30 vs B30 city investors behave differently. This project analyses 32,778 investor transactions to produce demographic and geographic insights.

Problem 5 — Reports are Slow and Static

Monthly MF reports are prepared manually and are static PDFs. By the time a stakeholder reads them, the data is already outdated. This project builds a Power BI dashboard that can be refreshed directly from the ETL pipeline output — making reporting fast and self-service.

PROJECT OBJECTIVE

The main goal of this project was to build a complete end-to-end Mutual Fund Analytics Platform using real publicly available data from AMFI India. starting from raw data files, cleaning them, storing them in a proper database, analysing them deeply, and finally presenting the findings in a way that anyone can understand.

First build an automated ETL pipeline ,then design a proper SQL database using a star schema structure. Instead of keeping everything in flat CSV files, the goal was to organise the data into a relational database that supports fast queries and can scale as more data is added. Thorough Exploratory Data Analysis. Compute industry-standard performance and risk metrics for all 40 funds.Then build an interactive Power BI dashboard .

The 10 Datasets Used

#	Dataset Name	Rows	What it Contains
01	fund_master.csv	40	Basic info on 40 fund schemes — AMC name, category, expense ratio, risk grade
02	nav_history.csv	46,000	Daily NAV for all 40 funds from January 2022 to May 2026
03	aum_by_fund_house.csv	90	Quarterly AUM for 10 fund houses over 4 years
04	monthly_sip_inflows.csv	48	Monthly SIP inflow amounts, active accounts, new registrations
05	category_inflows.csv	144	Net inflows by category — Large Cap, Mid Cap, Small Cap, ELSS, etc.
06	industry_folio_count.csv	21	Total mutual fund folios broken into Equity, Debt, Hybrid
07	scheme_performance.csv	40	Returns, Sharpe, Sortino, Alpha, Beta, Max Drawdown for each fund
08	investor_transactions.csv	32,778	SIP, Lumpsum, Redemption records for 5,000 investors
09	portfolio_holdings.csv	322	Top stocks held by each equity fund and their sector weights
10	benchmark_indices.csv	8,050	Daily Nifty 50, Nifty 100, Nifty Midcap 150, BSE SmallCap values

6. SYSTEM ARCHITECTURE AND ETL PIPELINE

What ETL Means

ETL It is the backbone of any data engineering project.

Extract means collecting raw ingredients, Loaded all 10 CSV files using Python Pandas. Fetched live NAV data from mfapi.in REST API for 5 schemes.

Transform means Parsed all date columns to datetime. Removed duplicate rows. Forward filled missing NAV values for weekends and holidays. Standardised transaction types. Validated AMFI codes across all datasets.

Load means Saved all cleaned files to processed folder. Loaded all 8 tables into SQLite database. Ran 10 SQL queries to verify data integrity.

The Full Architecture — 5 Layers

Layer 1 — Extract (Getting the Data)

- Loaded all 10 CSV datasets using Python and Pandas
- Fetched live NAV data from the mfapi.in REST API for 5 schemes using the requests library
- Verified that all file paths, AMFI codes, and date ranges were consistent

Layer 2 — Transform (Cleaning and Preparing the Data)

- Converted all date columns from plain text strings to proper Python datetime format
- Forward-filled missing NAV values for weekends and public holidays — because markets are closed on these days but the NAV record still needs to exist for continuity
- Removed duplicate rows from the NAV history dataset
- Standardised the transaction_type column — values like 'sip', 'SIP', 'Sip' were all converted to a consistent 'SIP' format
- Validated that all 40 AMFI codes in fund_master.csv existed in nav_history.csv — zero mismatches were found
- Computed derived columns like daily_return_pct directly during the transform stage so they don't need to be recalculated at query time

Layer 3 — Load (Saving to Database)

- All 8 cleaned tables were loaded into a SQLite database called bluestock_mf.db using SQLAlchemy
- Indexes were created on amfi_code and date columns for faster query performance
- CSV flat files were also saved to the processed folder for direct Power BI import

Layer 4 — Analyse (Running the Analytics)

- Exploratory Data Analysis performed in Jupyter Notebooks using Matplotlib, Seaborn, and Plotly
- Performance metrics computed using NumPy and SciPy
- Benchmark comparison done using OLS regression from scipy.stats

Layer 5 — Visualise (The Dashboard)

- 4-page interactive Power BI dashboard built from the processed data
- Slicers, KPI cards, drill-through, and tooltips added

3 Data Cleaning

Dataset	Problem Found	How It Was Fixed
nav_history.csv	Missing NAV on weekends and holidays	Forward-filled using ffill after reindexing to full date range
nav_history.csv	Duplicate date entries	Removed using drop_duplicates
investor_transactions.csv	Inconsistent date formats	Converted all to YYYY-MM-DD standard
investor_transactions.csv	Mixed transaction type values (sip/SIP/Sip)	Standardised all to SIP / Lumpsum / Redemption
scheme_performance.csv	Return columns stored as strings	Converted to float using pd.to_numeric
All datasets	AMFI code consistency check	Validated — 0 mismatches found across all databases

SQL Database Design — Star Schema

A star schema was chosen for the database because it is the industry standard for analytical databases. In a star schema there is one central fact table (storing numbers and events) surrounded by dimension tables (storing descriptions and labels). This makes queries fast and the structure easy to understand.

The 8 tables created:

Table	Type	What it Stores	Rows
dim_fund	Dimension	Master details of all 40 funds — AMC, category, expense ratio, risk grade	40
dim_date	Dimension	Every date from Jan 2022 to May 2026 with year, month, quarter, trading day flag	~1,500
fact_nav	Fact	Daily NAV and daily return % for all 40 funds	46,000
fact_transactions	Fact	Every SIP, Lumpsum and Redemption transaction	32,778
fact_performance	Fact	All pre-computed metrics — Sharpe, Alpha, Beta, CAGR, VaR — per fund	40
fact_portfolio	Fact	Top stock holdings and sector weights per equity fund	322
fact_aum	Fact	Quarterly AUM per fund house	90
fact_sip_industry	Fact	Monthly industry SIP inflow and account data	48

What EDA Is

Exploratory Data Analysis means looking at the data visually and statistically to find patterns, trends, and interesting insights before doing any complex modelling. I created 15+ charts across four areas — NAV trends, AUM growth, SIP inflows, and investor demographics.

NAV Trend Analysis (2022–2026)

Daily NAV for all 40 schemes was plotted from January 2022 to May 2026. The chart clearly shows three distinct market phases:

- **2022 Correction:** All funds dropped during the global market correction caused by US Federal Reserve interest rate hikes. Large cap funds fell less than mid and small cap funds.
- **2023 Rally:** Mid cap and small cap funds began significantly outperforming large cap funds during this period, driven by domestic consumption and infrastructure spending.
- **2024–2025 Bull Run:** Almost all equity funds delivered strong returns. Large cap funds tracked the Nifty 100 closely while small cap funds showed higher volatility.
- Axis Bluechip recorded the highest absolute NAV at Rs. 6,168.48 while SBI Bluechip showed the most consistent and smooth growth trajectory.

AUM Growth by Fund House (2022–2025)

- SBI Mutual Fund is the dominant player by a large margin. At Rs. 12.50 lakh crore, SBI MF manages nearly double the AUM of ICICI Prudential (Rs. 10.74 lakh crore). This is not because SBI's funds perform dramatically better — it is because SBI has the largest banking distribution network in India with over 22,000 branches, making it the most accessible AMC for retail investors across both urban and rural India.
- All 10 fund houses showed consistent AUM growth across 2022–2025. The total industry AUM doubled over this period, reflecting both new investor inflows and market appreciation.

SIP Inflow Trend (January 2022 – December 2025)

Monthly SIP inflows grew from Rs. 11,517 crore in January 2022 to Rs. 31,002 crore in December 2025. That is a growth of 169% in just 4 years. The growth was not uniform — it accelerated sharply from mid-2023 onwards, driven by SEBI-led investor education campaigns, the rise of mobile investing platforms, and growing financial awareness among younger Indians.

The all-time high of Rs. 31,002 crore in December 2025 is particularly significant because it shows that SIPs are no longer a niche product — they have become the dominant way retail India invests in equity markets.

Investor Age Group Analysis

Age Group	Share of Total Investors
26–35 years	41.1%
36–45 years	24.9%
18–25 years	15.0%
46–55 years	11.5%
56+ years	7.5%

- The 26-35 age group is the largest SIP investor segment.

Geographic Distribution

Maharashtra, Karnataka, and Delhi lead in SIP investment volumes — consistent with their status as India's top financial, tech, and government hubs. T30 cities (Top 30 cities as classified by AMFI) contribute over 70% of total investment amount while B30 cities (Beyond Top 30) contribute the remaining 30%.

This 70-30 split shows there is massive untapped potential in smaller cities and towns. AMFI has specific incentives for AMCs to push distribution into B30 cities, and this data confirms those cities are growing but still significantly behind.

Folio Count Growth

Total investor folios doubled from 13.26 crore in early 2022 to 26.12 crore by December 2025. This is a 4-year CAGR of approximately 18.5%. Equity folios drove almost all of this growth, consistent with the bull market and the rising SIP culture.

Sector Allocation Across Equity Portfolios

Looking at the combined stock holdings of all equity funds, Banking is the largest sector at 19.2% followed by IT at 13.4% and Pharma at 12.0%:

Sector	Portfolio Weight
Banking	19.2%
IT	13.4%
Pharma	12.0%
Automobile	9.5%
Utilities	7.8%
FMCG	6.7%
Infrastructure	5.7%
Diversified	5.0%
Others	20.7%

Banking and IT together account for over 32% of all equity fund portfolios. This is important because it means all large cap equity funds carry significant concentration risk in these two sectors. A banking sector slowdown or an IT industry downturn would simultaneously hit almost every large cap equity fund.

Correlation Analysis

The pairwise correlation of daily NAV returns across large cap funds shows extremely high correlation — all above 0.85. This means that if one large cap fund goes up by 1%, all other large cap funds also go up by roughly 0.85% or more on the same day. They essentially move together.

This has a very practical implication for investors — holding SBI Bluechip, HDFC Top 100, and Axis Bluechip together does NOT reduce risk in any meaningful way. They are not truly diversified. Genuine diversification requires combining large cap funds with mid cap, small cap, or debt funds.

FUND PERFORMANCE ANALYTICS

8.1 How Returns Were Calculated

Daily Return: The percentage change in NAV from one day to the next.

Formula: $(\text{NAV today} / \text{NAV yesterday}) - 1$

CAGR (Compounded Annual Growth Rate): The annualised return over 1 year, 3 years, and 5 years.

Formula: $(\text{NAV at end} / \text{NAV at start})^{(1 / \text{number of years})} - 1$

All returns were annualised using 252 trading days per year — the standard in the finance industry.

8.2 Risk Metrics — What Each One Means

Sharpe Ratio

This is the most widely used measure of risk-adjusted return. It tells you how much return the fund earned per unit of risk taken. A Sharpe Ratio above 1 is generally considered good.

Sortino Ratio

Similar to Sharpe but smarter — it only penalises the fund for downside volatility (the bad days), not total volatility. A fund that is volatile on the upside shouldn't be punished the same way as one that is volatile on the downside.

Alpha

Alpha tells you how much extra return the fund delivered above what its level of market exposure would predict. Positive alpha means the fund manager added genuine value beyond just riding the market.

Computed by regressing fund returns against Nifty 100 returns using OLS (Ordinary Least Squares) regression.

Beta

Beta measures how sensitive the fund is to market movements. A Beta of 1 means the fund moves exactly with the market. Beta of 1.2 means if the market falls 10%, this fund is likely to fall 12%. Computed as the slope of the OLS regression line.

Maximum Drawdown

The worst fall from a peak before the NAV recovered. This shows an investor the worst-case loss they could have experienced if they invested at the peak and sold at the bottom.

Value at Risk (VaR) at 95% Confidence

VaR answers the question — on a typical bad day, how much can this fund lose? At 95% confidence, it means there is only a 5% chance of losing more than this amount on any given day.

Results from VaR Analysis:

- Small cap funds: VaR of -2% to -3% per day
- Large cap funds: VaR of -1% to -1.5% per day

This confirms that small cap funds carry significantly more short-term risk than large cap funds, even though their long-term returns can be higher.

Fund Scorecard — Composite Ranking

Instead of ranking funds purely by return (which rewards funds that took excessive risk), a composite scorecard was built that considers five factors:

Factor	Weight	Reason
3-Year Return Rank	30%	Most important — shows consistent medium-term performance
Sharpe Ratio Rank	25%	Rewards funds that earned returns efficiently
Alpha Rank	20%	Rewards fund managers who genuinely beat the benchmark
Expense Ratio Rank (inverse)	15%	Lower fees benefit investors — lower expense ratio gets a higher rank
Max Drawdown Rank (inverse)	10%	Smaller drawdowns = lower risk — lower drawdown gets a higher rank

This composite score gives a much more balanced and honest picture of fund quality than any single metric alone. A fund that scores high on this composite scorecard is one that delivered good returns, managed risk well, beat the benchmark, kept costs low, and protected investors during market downturns.

ADVANCED ANALYTICS

Instead of looking at Sharpe Ratio as a single fixed number, the rolling 90-day Sharpe was computed — meaning the Sharpe Ratio was recalculated every day using only the last 90 days of returns. This gives a dynamic picture of how risk-adjusted returns have changed over time.

- All 5 large cap funds dipped below Sharpe = 1 during the 2022 global correction
- By 2024, Sharpe ratios had recovered to above 1.2 for most funds
- The rolling Sharpe chart is useful for investors to identify whether a fund is in a good period or a weak period — helping them time additional investments

Investor Cohort Analysis

Investors were grouped by the year of their first transaction — 2022, 2023, 2024, or 2025 — and compared on average SIP amount, total invested, and fund preferences:

- The 2024 cohort invests significantly higher SIP amounts on average compared to the 2022 cohort
- Later cohorts show a stronger preference for Direct plans over Regular plans
- This suggests that financial literacy among new investors is improving — they are more aware of expense ratios and the long-term benefits of Direct plans

SIP Continuity Analysis

For every investor with 6 or more SIP transactions on record, the average number of days between consecutive SIP payments was calculated. Investors with gaps of more than 35 days between SIP payments were flagged as "At Risk" of discontinuing their SIP.

Fund Recommendation Engine

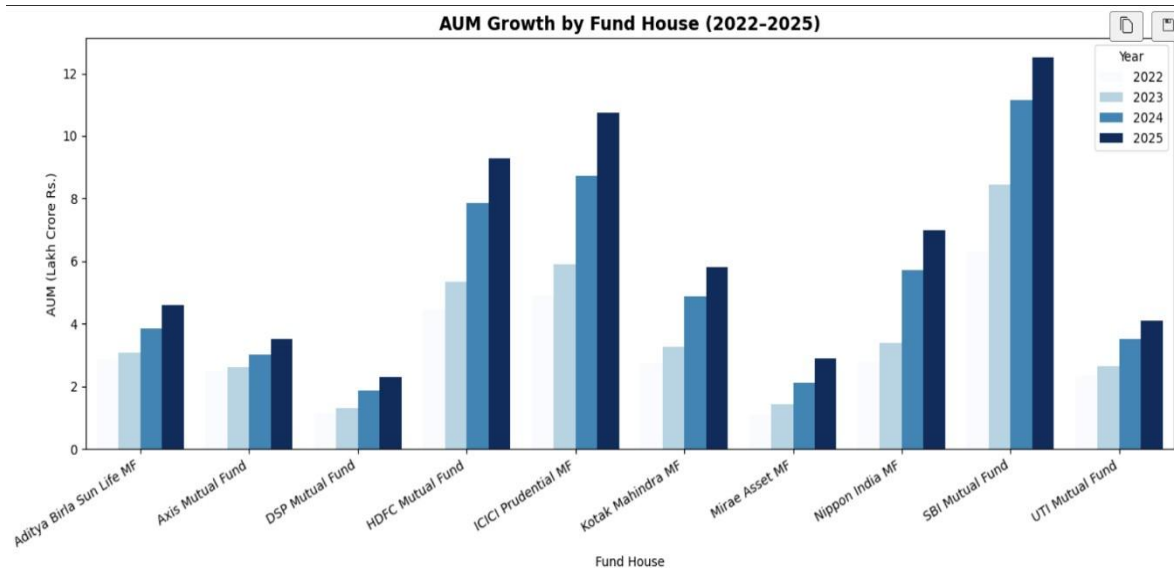
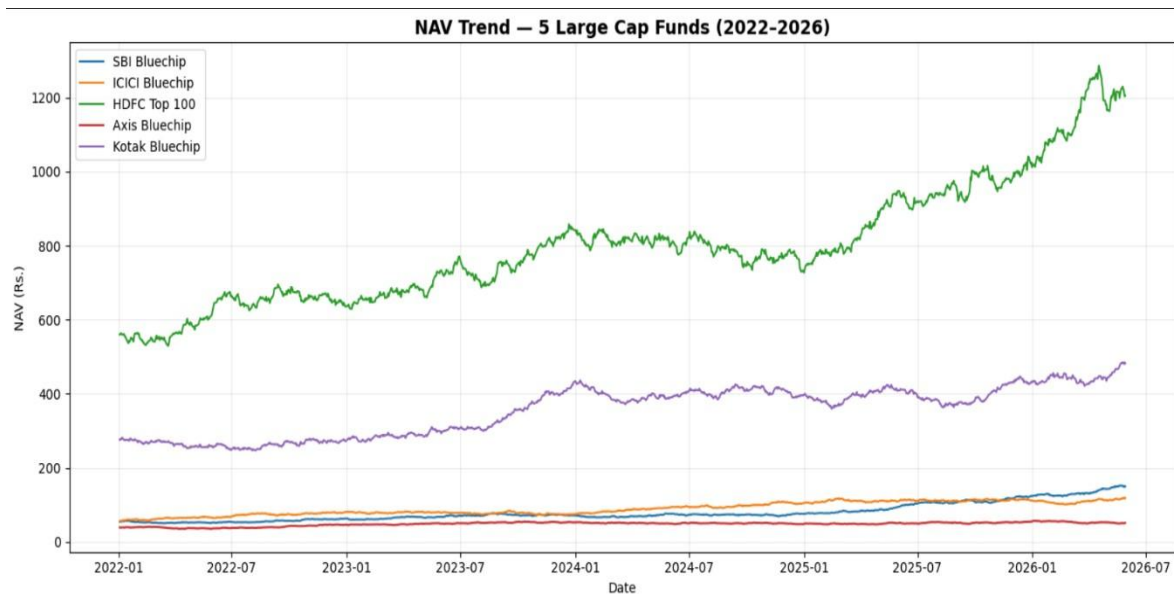
A simple rule-based fund recommendation system was built in Python:

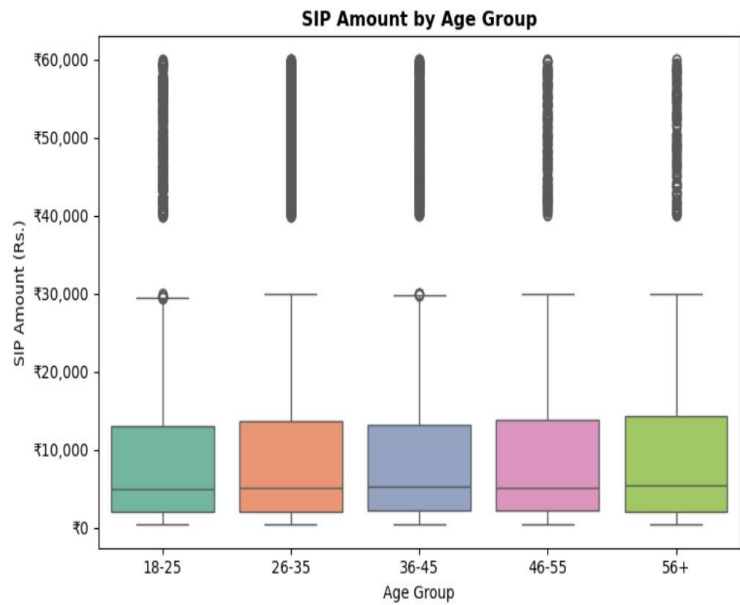
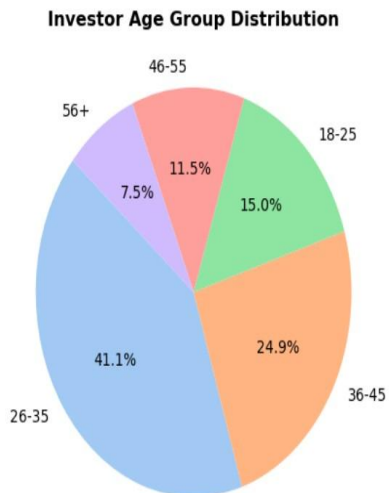
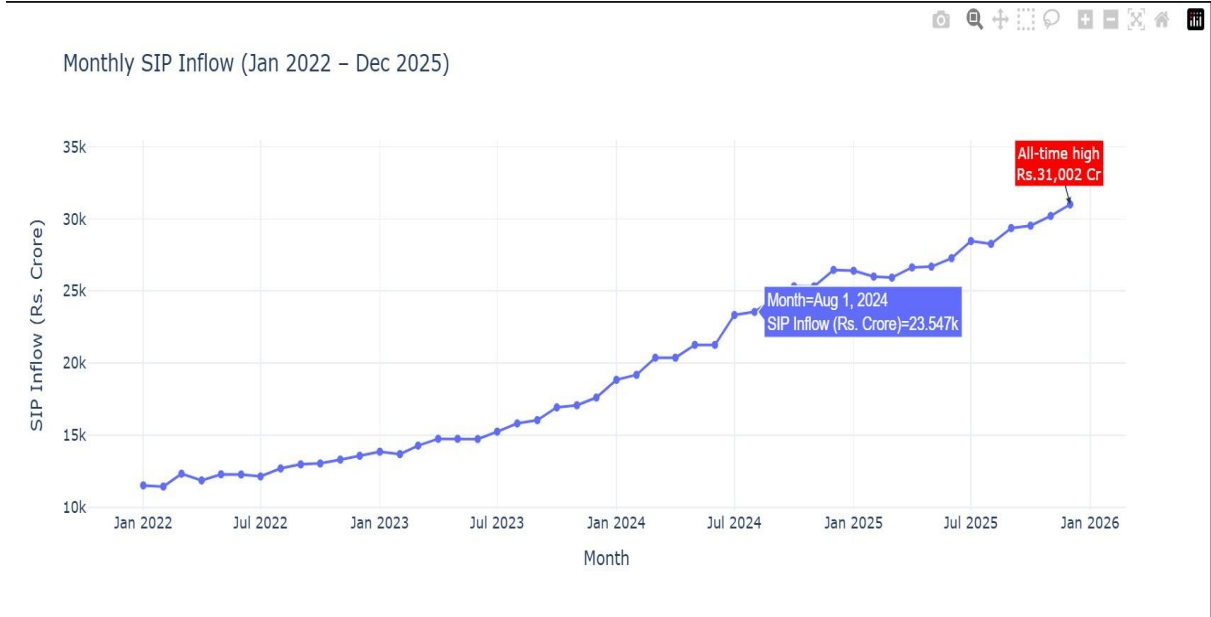
- Input: Investor's risk appetite — Low, Moderate, or High
- Logic: Filter the 40 funds by matching SEBI risk grade, then rank by Sharpe Ratio within that group
- Output: Top 3 recommended funds for the investor's risk profile

Sector Concentration Risk — HHI Analysis

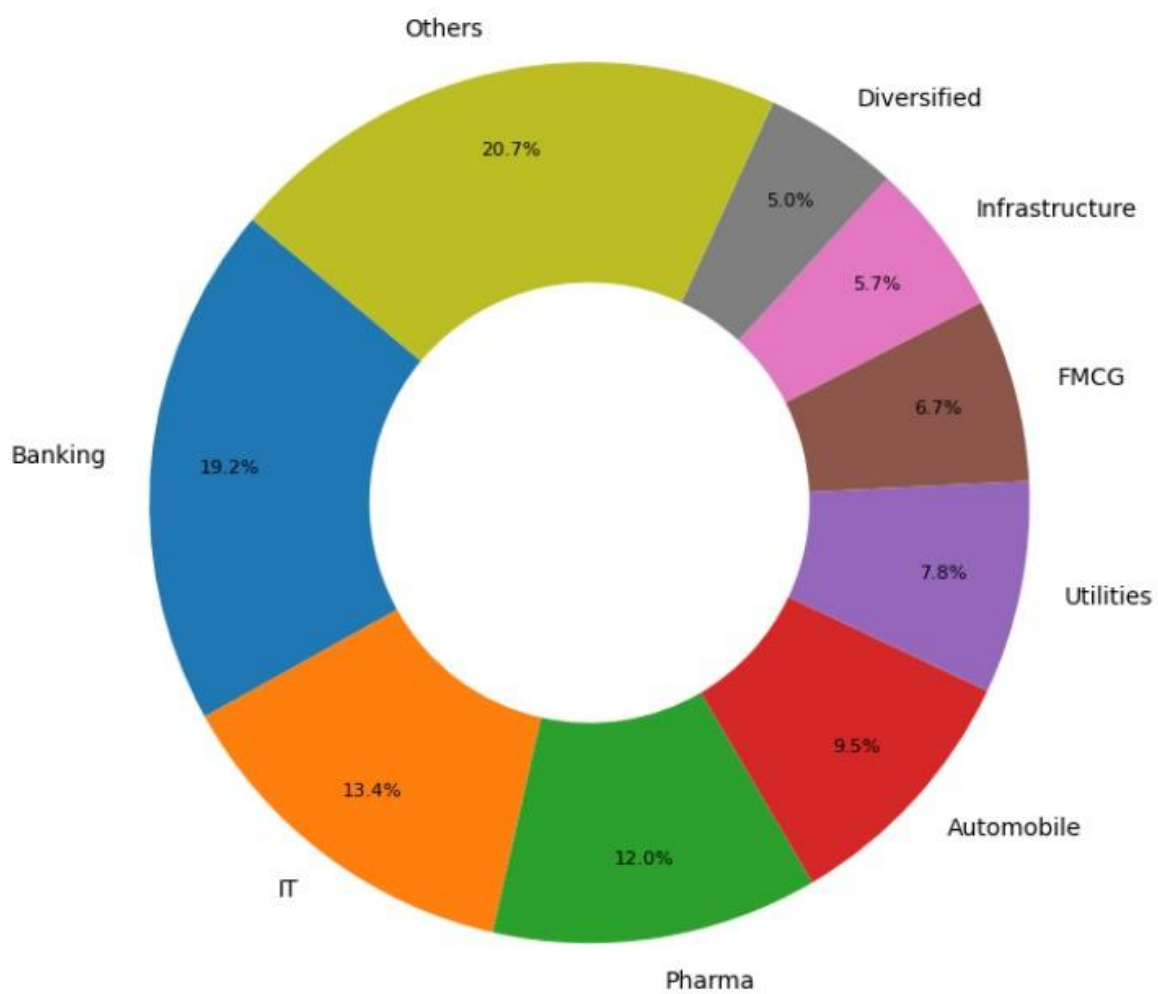
For each equity fund, the Herfindahl-Hirschman Index (HHI) was calculated using the sector weights of the fund's portfolio: $HHI = \text{sum of (each sector weight squared)}$

A higher HHI means the fund is more concentrated in fewer sectors — which means higher sector-specific risk. This analysis revealed that some funds appearing to be well-diversified by name were actually heavily concentrated in Banking and IT, which are the two dominant sectors in the Indian large cap universe.

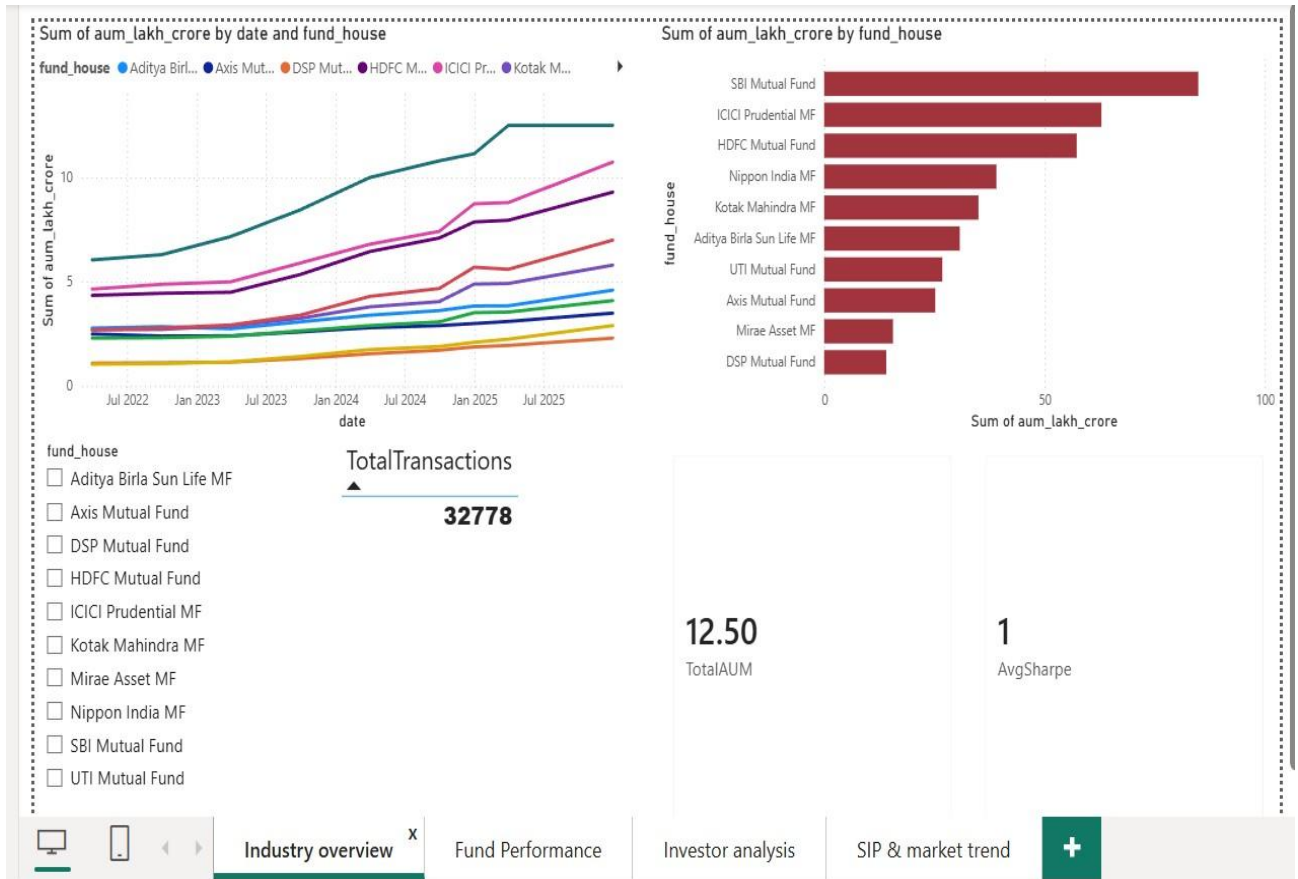




Sector Allocation Across Equity Fund Portfolios



Power BI Dashboard

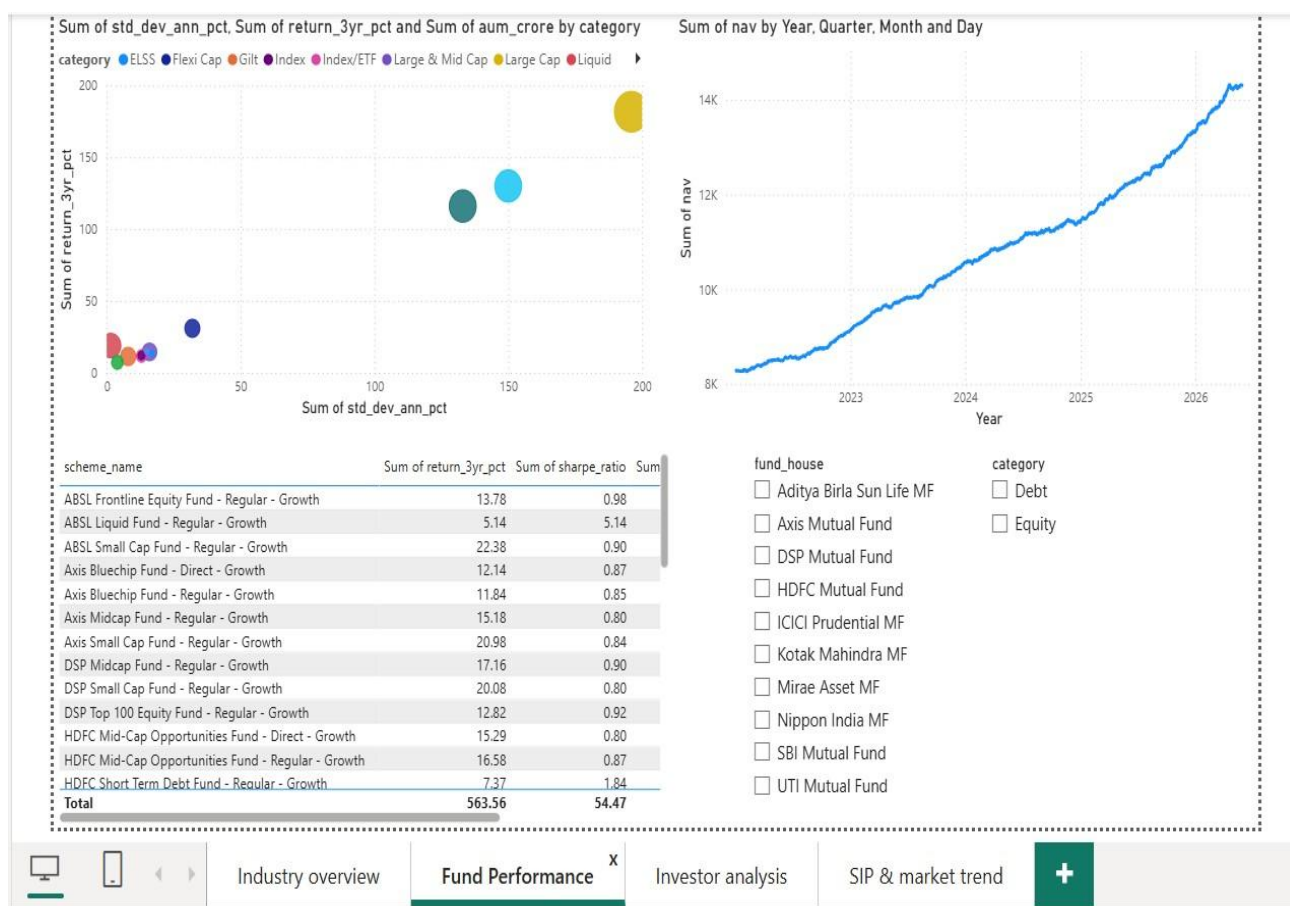


1. Industry Overview

Designed for: Senior management, fund managers

What it shows:

- KPI cards — Total AUM (Rs. 12.50 lakh crore), Total Transactions (32,778), Average Sharpe Ratio (1.0)
- Line chart showing AUM trend from July 2022 to July 2025 across all fund houses
- Horizontal bar chart showing AUM by fund house — SBI MF clearly dominates
- Slicer to filter by fund house

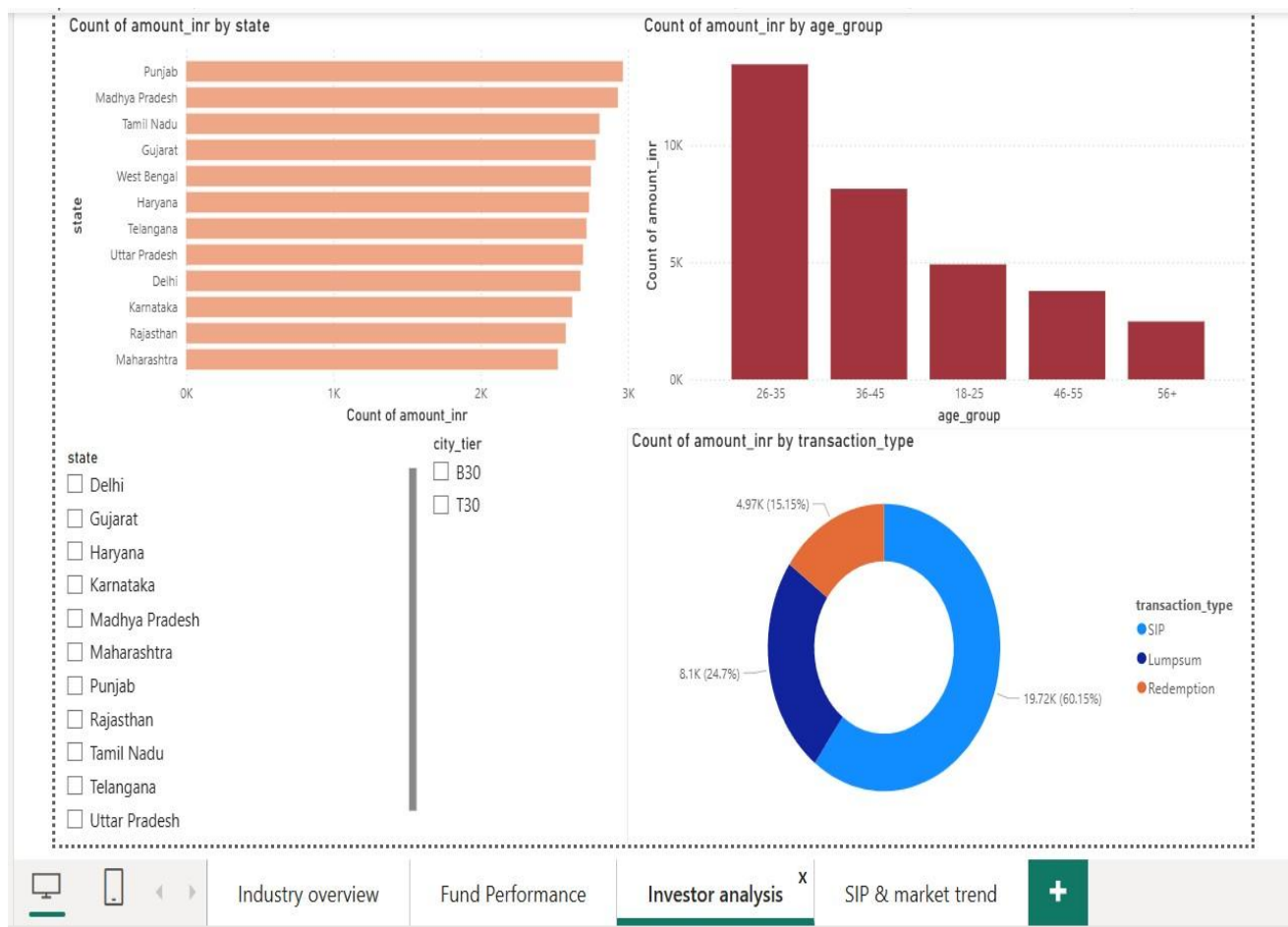


2. Fund Performance

Designed for: Investment analysts, financial advisors

What it shows:

- Scatter plot with Risk (standard deviation) on X-axis and 3-year return on Y-axis — each bubble is a fund
- Sortable fund scorecard table showing all 40 funds with their 3yr return, Sharpe Ratio, and Alpha side by side
- NAV history line chart that updates dynamically when a fund is selected from the scorecard table
- Slicers for Fund House, Category (Equity/Debt), and Plan (Direct/Regular)

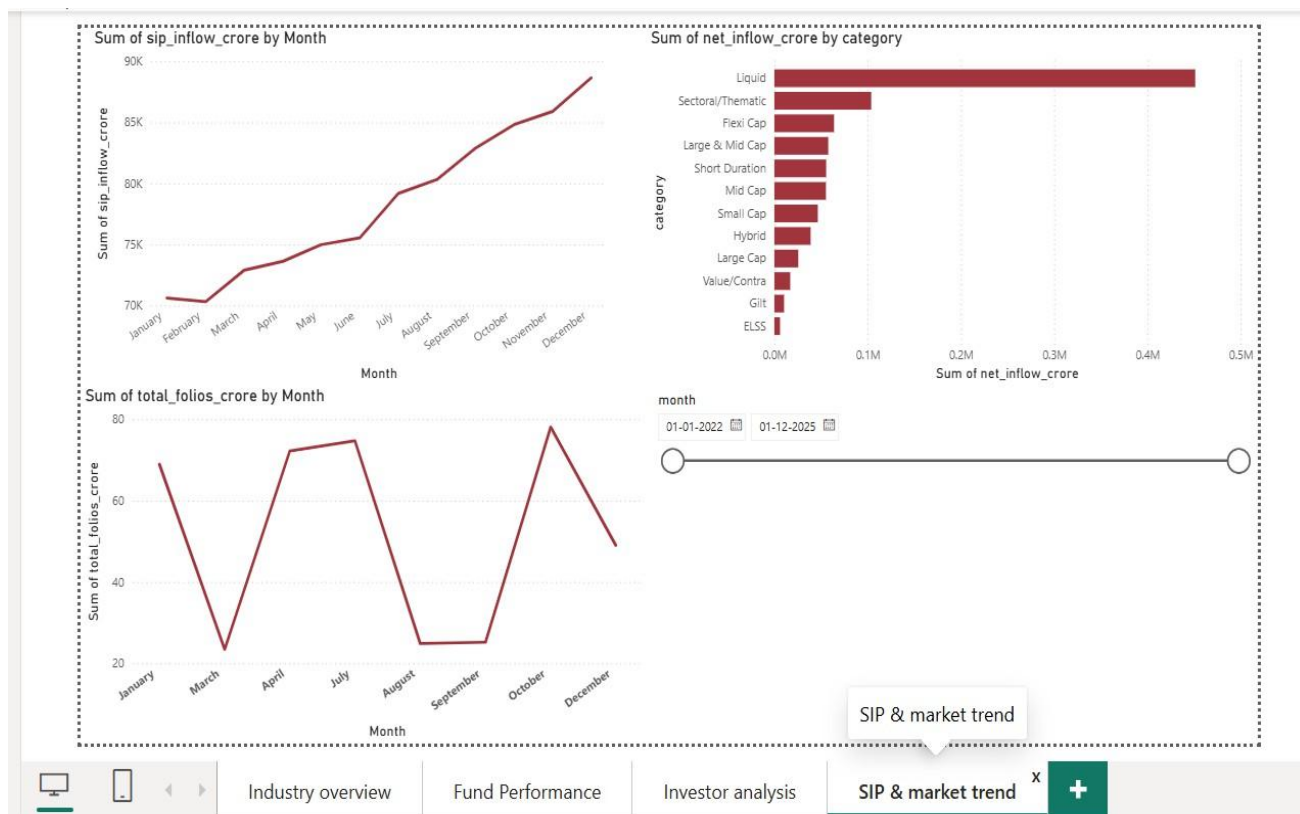


3. Investor Analytics

Designed for: Distribution teams, sales managers

What it shows:

- Horizontal bar chart of transaction counts and amounts by state — Maharashtra, Karnataka, Delhi at the top
- Donut chart showing transaction type split — SIP 60.15%, Lumpsum 24.7%, Redemption 15.15%
- Column chart showing SIP amounts by age group — 26-35 clearly leading
- T30 vs B30 city tier pie chart showing the 70-30 urban-rural investment split



4. SIP and Market Trends

Designed for: Strategy teams, product managers, regulators

What it shows:

- Monthly SIP inflow trend line from January 2022 to December 2025 — showing the steady climb to the all-time high of Rs. 31,002 crore
- Category net inflows bar chart showing which fund categories (Sectoral, Flexi Cap, Large & Mid Cap) attracted the most money in FY25
- Folio count growth line chart tracking the doubling of investor folios from 13.26 to 26.12 crore.

LIMITATIONS

Simulated investor data: The 32,778 investor transaction records use realistic demographic and geographic distributions based on AMFI data, but individual investor records are synthetically generated. Real investor data is not publicly available due to privacy regulations.

40 schemes out of 1,908: The project covers a curated selection of 40 funds from 10 major AMCs. The Indian industry has 1,908 schemes — this platform covers only about 2% of all schemes. The methodology is sound but the coverage is limited.

Power BI Desktop dependency: The dashboard requires Power BI Desktop to open. Sharing it with stakeholders who don't have Power BI Desktop installed requires either publishing to Power BI Service (paid licence) or exporting to PDF (which loses interactivity).

Batch ETL — no live refresh: The pipeline currently runs manually. NAV data is only as current as the last time the pipeline was run. There is no automatic daily refresh.

Benchmark coverage: Only Nifty 50, Nifty 100, Nifty Midcap 150, and BSE SmallCap are included. Debt funds should be benchmarked against CRISIL indices but this comparison is not fully implemented yet.

During this project I found it interesting that SBI Mutual Fund has almost double the AUM of ICICI Prudential. I also noticed that younger investors in the 26-35 age group are investing more through SIP compared to older age groups. Building the ETL pipeline taught me how messy real world data can be and the Power BI dashboard helped me understand how to present data to non technical audiences.

CONCLUSION

This project successfully built a complete end-to-end Mutual Fund Analytics Platform — taking raw publicly available data and turning it into a professional, insight-driven Power BI dashboard. All 8 project objectives were completed within the 7-day timeline.

The most important lesson from this project is that raw return numbers alone are not enough to evaluate a mutual fund. A fund that returned 30% in one year but had a Sharpe Ratio of 0.6 and a Maximum Drawdown of -40% is a fundamentally different and much riskier product than a fund that returned 22% with a Sharpe of 1.3 and a Maximum Drawdown of -18%. Risk-adjusted metrics tell a much more complete and honest story, and this platform makes all those metrics accessible in one place.

Building the ETL pipeline was the most technically challenging part of the project — handling inconsistent date formats, missing NAV values, and duplicate records taught me how messy real-world data can be even when it comes from an official source like AMFI. The Power BI dashboard section taught me how critical it is to present data in a way that non-technical stakeholders can understand and act on — the same numbers in a raw CSV table vs an interactive visual dashboard generate completely different levels of engagement and decision-making quality.

The Indian mutual fund industry is at an inflection point. With 26 crore investor folios, Rs. 81 lakh crore in AUM, and SIP inflows hitting Rs. 31,002 crore per month, the growth is real and structural. Platforms like this one — that make fund data transparent, comparable, and accessible — are exactly what the industry needs to help the next 100 crore Indians make better investment decisions.